

SPRING 2025 (CSE423 QUIZ)

Time: 30 Minutes

Name: _____ ID: _____

Section: _____

A

[Question 1 is mandatory. Answer any one from 2 - 3]

1. Given four control points $P_0=(0,1)$, $P_1=(1,2)$, $P_2=(3,-1)$, and $P_3=(4,0)$, find the point on the **cubic Bézier curve** at $t=0.2$. [8]
2. Given two **cubic Bézier curves** with 4 control points as follows: [6+6]
 - Curve A Control points: $A_0=(0,0)$, $A_1=(1,3)$, $A_2=(2,3)$, $A_3=(3,0)$
 - Curve B Control points: $B_0=?$, $B_1=?$, $B_2=?$, $B_3=?$

You have decided to join them together to form a closed loop (Closed loop means the starting of curve B will join with the ending of curve A and the ending of curve B will join with the starting of curve A similar to a circle or ellipse). Now find the control points of curve B in the following two scenarios:

 - a. At both joining points, the curve will be C(1) continuous.
 - b. At both joining points, the curve will be G(1) continuous but will not be C(1) continuous.
3. You have a **cubic Bézier curve (A)** with 4 control points: $A_0=(0,0)$, $A_1=(3,1)$, $A_2=(3,2)$, $A_3=(0,3)$. But you wanted to end at the point (7, 6). So you wanted to join another **cubic Bézier curve (B)** at the end of **A** such that the new curve ends at (7, 6). [$B_3 = (7, 6)$]
However, you want to make sure the curve is very smooth which is why you want to make sure **C(2)** continuity at the joining point. **Now, determine all the control points of curve B.** [12]
Hint: Make sure, $A(1) = B(0)$, $A'(1) = B'(0)$ and $A''(1) = B''(0)$

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[Question 1 is mandatory. Answer any one from 2 - 3]

B

1. Given four control points $P_0=(-1,0)$, $P_1=(0,1)$, $P_2=(2,-2)$, and $P_3=(3,-1)$, find the point on the **cubic Bézier curve** at $t=0.2$. [8]
2. Given two **cubic Bézier curves** with 4 control points as follows: [6+6]
 - Curve A Control points: $A_0=(0,0)$, $A_1=(1,3)$, $A_2=(2,3)$, $A_3=(3,0)$
 - Curve B Control points: $B_0=?$, $B_1=?$, $B_2=?$, $B_3=?$

You have decided to join them together to form a closed loop (Closed loop means the starting of curve B will join with the ending of curve A and the ending of curve B will join with the starting of curve A similar to a circle or ellipse). Now find the control points of curve B in the following two scenarios:

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However, you want to make sure the curve is very smooth which is why you want to make sure **C(2)** continuity at the joining point. **Now, determine all the control points of curve B.** [12]
Hint: Make sure, $A(1) = B(0)$, $A'(1) = B'(0)$ and $A''(1) = B''(0)$

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[Question 1 is mandatory. Answer any one from 2 - 3]

C

1. Given four control points $P_0=(1,2)$, $P_1=(2,3)$, $P_2=(4,0)$, and $P_3=(5,1)$, find the point on the **cubic Bézier curve** at $t=0.2$. [8]
2. Given two **cubic Bézier curves** with 4 control points as follows: [6+6]
 - Curve A Control points: $A_0=(0,0)$, $A_1=(1,3)$, $A_2=(2,3)$, $A_3=(3,0)$
 - Curve B Control points: $B_0=?$, $B_1=?$, $B_2=?$, $B_3=?$

You have decided to join them together to form a closed loop (Closed loop means the starting of curve B will join with the ending of curve A and the ending of curve B will join with the starting of curve A similar to a circle or ellipse). Now find the control points of curve B in the following two scenarios:

- a. At both joining points, the curve will be C(1) continuous.
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However, you want to make sure the curve is very smooth which is why you want to make sure **C(2)** continuity at the joining point. **Now, determine all the control points of curve B.** [12]
Hint: Make sure, $A(1) = B(0)$, $A'(1) = B'(0)$ and $A''(1) = B''(0)$